Set Theory Foundations & Elementary Proofs

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Problem 1: De Morgan's Law Proof

Theorem 1. For any sets A and B, $(A \cup B)^c = A^c \cap B^c$.

Proof. We show mutual inclusion $(A \cup B)^c \subseteq A^c \cap B^c$ and $A^c \cap B^c \subseteq (A \cup B)^c$.
Let $x \in (A \cup B)^c$ be arbitrary.

$$\begin{aligned}
x \in (A \cup B)^c &\iff x \notin (A \cup B) \\
&\iff \neg(x \in A \vee x \in B) \\
&\iff (x \notin A) \wedge (x \notin B) \\
&\iff (x \in A^c) \wedge (x \in B^c) \\
&\iff x \in (A^c \cap B^c)
\end{aligned}$$

Since every logical equivalence step holds bi-directionally, $(A \cup B)^c = A^c \cap B^c$. $\square$

Problem 2: Matrix & Distributive Property

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. The Cartesian product property holds as follows:

$$A \times (B \cap C) = (A \times B) \cap (A \times C) \tag{1}$$